

Gain-switched DOFC for real-time interrogation of FBGs

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Abstract *We demonstrate a real-time fibre Bragg grating interrogation system using externally injected gain-switched dual optical frequency comb and embedded receiver. The system achieves interrogation rates up to 1 kHz, while a minimum Bragg wavelength tracking uncertainty of 64.9 fm is obtained at reduced measurement rate.*

Introduction

Fibre Bragg gratings (FBGs) are widely used in optical sensing to measure physical parameters such as temperature [1] and strain [2], owing to their high sensitivity, compactness, and immunity to electromagnetic interference. Conventional FBG interrogation approaches based on broadband sources, such as amplified spontaneous emission (ASE), together with optical spectrum analysers (OSAs) [3], provide direct spectral measurement but are typically limited by low acquisition speed and bulky size. Tunable laser (TL)-based interrogation is another commonly used technique [4], where the wavelength of the laser is swept across the FBG spectral profile, while the reflected or transmitted signal is detected by a photodiode (PD) and synchronised with the instantaneous wavelength to reconstruct the spectral response. As a result, precise synchronisation and wavelength scanning are required, and true single-shot measurements cannot be achieved.

Dual optical frequency comb (DOFC)-based interrogation has emerged as a promising alternative for FBG sensing [5, 6]. In this approach, two optical frequency combs (OFC) with slightly different free spectral ranges (FSRs) are used, with a pair of comb lines (one from each OFC) is aligned to provide a common optical reference. Owing to the multi-wavelength nature of OFCs, evenly spaced comb-line pairs act as discrete sampling points across the FBG spectrum, enabling single-shot interrogation without wavelength scanning. Subsequently, heterodyne detection between the comb-line pairs maps the optical spectral information into a radio-frequency (RF) comb with a tone spacing equal to the difference in FSR (ΔFSR), enabling the use of low-cost, low-frequency RF components for signal detection and processing. Various DOFC generation techniques have

been reported, including mode-locked laser (MLL)-based [5, 6], electro-optic modulator (EOM)-based approaches [7, 8], and externally injected gain-switched laser (EI-GSL)-based schemes [9]. Among these, the EI-GSL-based DOFC generation provides a simple and cost-effective architecture, which is amenable to be realised as a photonic integrated circuit (PIC) [10]. In addition, it also offers flexible wavelength and FSR control, all of which make it attractive for practical sensing applications.

In this work, we demonstrate a real-time FBG interrogation system based on an EI-GSL DOFC interrogator combined with a custom embedded receiver. The DOFC interrogator enables optical-to-RF mapping of the reflected spectrum, while the embedded receiver performs real-time acquisition and curve fitting on the down-converted RF comb to estimate the centre frequency (CF), which is directly related to the Bragg wavelength. This architecture allows direct extraction of the FBG response in the RF domain without wavelength scanning, making it attractive for static and dynamic sensing. Real-time processing implemented on the embedded platform enables interrogation rates of up to 1 kHz. Static interrogation experiments demonstrate a minimum interrogation uncertainty of 64.9 fm. To the best of our knowledge, this represents the lowest reported interrogation uncertainty among DOFC-based FBG interrogation systems, while simultaneously achieving one of the highest interrogation rates [2, 11, 12].

System Architecture

The setup of the interrogation system is shown in Fig. 1. The EI-GSL-based DOFC is used as the interrogator, while the PD and the custom embedded platform form the receiver. This interrogator has a primary–secondary laser configuration, where a narrow-linewidth TL is

used as the primary laser (PL), while two gain-switched Fabry–Perot (FP) lasers are used as the secondary lasers, each driven by a sinusoidal electrical signal, as illustrated by the dashed blue box in Fig. 1.

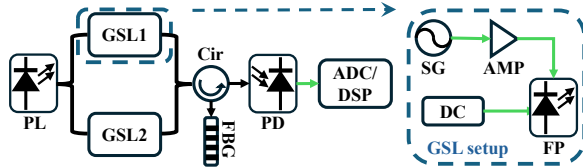


Fig. 1: Experimental setup of the FBG interrogation system, PL (primary laser), GSL (gain-switched laser), Cir (circulator), PD (photodetector), SG (RF signal generator), AMP (RF amplifier), ADC (analog-to-digital converter).

The optical spectra of the two OFCs are shown in Fig. 2. The output of the single PL, serving as the reference wavelength for both OFCs, injection locks both secondary lasers. The two FP lasers are modulated at slightly different frequencies (1 GHz and 999.98 MHz, respectively), generating two OFCs with a small difference in FSR. These two combs are then combined and directed towards the FBG. The reflected signal from the FBG is subsequently down-converted to the RF domain through heterodyne detection.

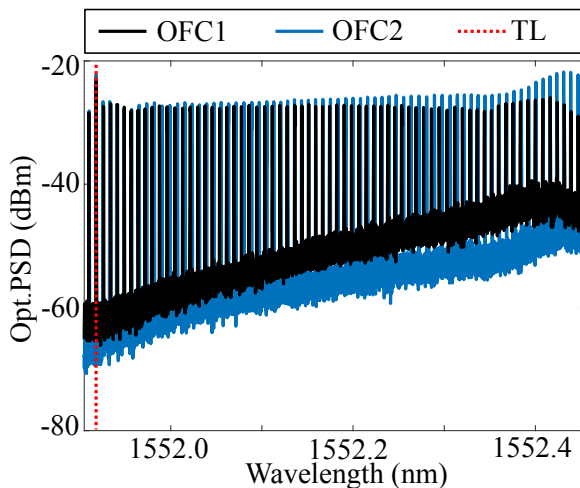


Fig. 2: Optical spectra of the generated OFCs with the same external injection.

The PD converts the reflected optical signal into an electrical waveform, which is then fed into the embedded platform. The platform comprises an analog front-end, an analog-to-digital converter (ADC, ADC14L029), and a microcontroller unit (MCU, i.MXRT1060 600 MHz ARM Cortex M7), integrated into a compact processing unit. The analog front-end provides low-distortion signal conditioning prior to digitisation, while the ADC operates at a 10 MHz sampling rate with a low-jitter clock. This enables accurate acquisition of the reflected RF comb using a compact and cost-effective embedded hardware solution. A schematic of the hardware and software processing pipeline of the real-time ADC/DSP unit is shown in Fig. 3.

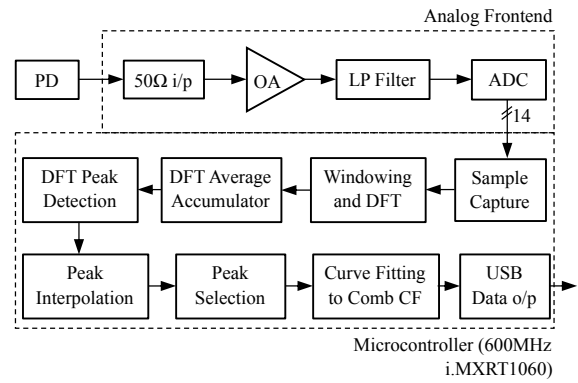


Fig. 3: Structure and processing pipeline of the real-time ADC/DSP unit. PD (photodiode), OA (operational amplifier), LP filter (low-pass filter) ADC (analog-to-digital converter), DFT (discrete Fourier transform).

The processing chain starts with sample capture of the PD output, followed by windowing and discrete Fourier transform (DFT) computation. DFT averaging is then applied to reduce frequency-domain noise. The number of samples and DFT averaging factor can be adjusted to balance measurement rate and resolution. The spectral peaks are subsequently detected and interpolated to improve the estimation accuracy of the RF comb-line parameters. Finally, curve fitting is performed to estimate the CF of the reflected RF comb. By implementing the complete ADC/DSP chain on the embedded platform, the system can perform real-time interrogation without off-line post-processing.

Experiment and Results

To improve the measurement resolution, the two OFCs were first characterised and optimised according to an OFC figure of merit (FoM) [10] to enhance comb quality and reduce RF amplitude fluctuations, thereby improving the stability of CF tracking and reducing the resulting interrogation uncertainty. The optimised OFCs exhibit an overlapping 3-dB spectral window of approximately 0.43 nm, with side-mode suppression ratios (SMSRs) of 26.23 dB and 26.10 dB, and average SNR (signal to noise ratio) of 22.88 dB and 28.53 dB for OFC 1 and OFC 2, respectively. It is important to note that an optical injection-based trade-off exists between comb flatness and SMSR. Improving the SMSR typically reduces the usable spectral bandwidth [13], which in turn limits the effective interrogation range. Following OFC optimisation, a 30-minute amplitude stability test was carried out within the selected 3-dB spectral window, where all comb lines exhibited amplitude fluctuations of less than 0.9 dB.

After OFC optimisation, the system was used for FBG interrogation. A commercial tunable FBG is used as the device under test, whose Bragg wavelength can be precisely tuned. A representative example of the reflected and

detected RF comb is shown in Fig. 4, where the comb envelope follows the reflected FBG profile, and a curve fitting method, we refer to as DualPoly4, is implemented on the embedded receiver for real-time CF estimation. In DualPoly4, the peak points used for fitting are first selected from the detected RF tones within a predefined amplitude window relative to the maximum peak. In this work, the upper threshold is set to 1.5 dB below the maximum peak, and a window width of 15 dB is used, allowing the exclusion of low-SNR tones while retaining the comb lines located on the edges of the FBG profile. An initial CF estimate is then obtained using a quadratic least-squares (Quad) fit, where a parabola is fitted to the selected peaks and the vertex is taken as the CF estimate. This estimate is used to divide the selected points into left- and right-hand subsets. A fourth-order polynomial is independently fitted to each subset, while the initial CF estimate is used as the fixed centre for both fits. Finally, each polynomial is intersected with the mid-window level, and the midpoint of the two intersection frequencies is taken as the estimated CF. This approach is well suited to the present interrogation system, as it enables accurate CF tracking for both symmetric and non-symmetric reflected RF comb profiles, while maintaining low computational overhead, with the curve fitting stage requiring only 0.208 ms per estimation, accounting for approximately 20% of the total processing time.

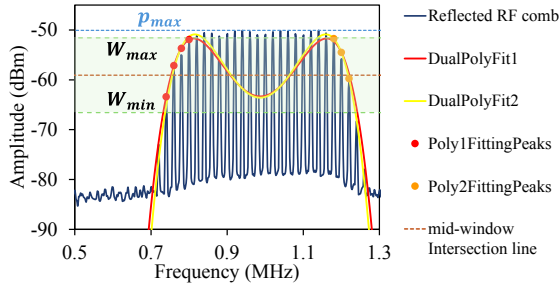


Fig. 4: RF comb spectra generated by the DOFC after reflection from the FBG, acquired with a record length of 4,096 samples and a DFT averaging factor of 50. Fitting using the DualPoly4 curve applied.

Following the CF estimation demonstrated in Fig. 4, static interrogation experiments were carried out using the optimised DOFC interrogator and the customised real-time processing platform running the DualPoly4 estimator. The Bragg wavelength of the FBG was adjusted to a set of discrete values such that the corresponding down-converted CFs in the RF domain were 405 kHz, 505 kHz, 605 kHz, 705 kHz, 805 kHz, 905 kHz, and 985 kHz. For each of these settings, the CF was recorded for 100 s using a sample length of 4096 and a DFT averaging factor of 500, yielding a measurement rate of 2.63 CF measurements per second. The standard

deviation (σ) of the estimated CF was taken as the measurement uncertainty.

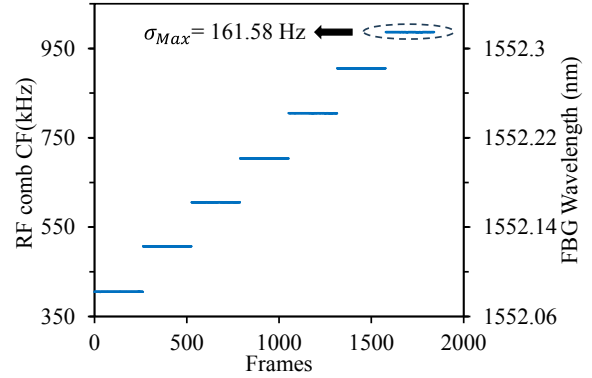


Fig. 5: Static FBG interrogation results Estimated CF for different Bragg-wavelength settings

The corresponding results are shown in Fig. 5. The estimated CF values shift consistently with the Bragg wavelength, confirming the correct mapping between the optical response of the FBG and its RF-domain representation. The CF estimates at the selected wavelength steps exhibit fluctuations in the order of hundreds of Hz. The largest variation was observed at 985 kHz, where the CF standard deviation was 161.58 Hz. The corresponding Bragg wavelength tracking uncertainty was calculated using equation (1):

$$\lambda_{\text{uncer}} = \frac{\text{STDEV}_{CF}}{f_{ls}} \times \text{FSR}, \quad (1)$$

where f_{ls} is the RF comb-line spacing and FSR is the optical comb spacing in wavelength. Based on this mapping, the maximum measured CF fluctuation corresponds to a Bragg wavelength tracking uncertainty of 64.9 fm.

Conclusion

A real-time FBG interrogation system based on an EI-GSL dual optical frequency comb interrogator and a custom embedded receiver has been demonstrated. By combining OFC optimisation and the DualPoly4 curve fitting method, the CF of the reflected RF comb, which is directly related to the Bragg wavelength, can be accurately tracked in the RF domain without wavelength scanning. The embedded receiver enables real-time acquisition and signal processing (on the RF comb), eliminating the need for off-line post-processing. Static interrogation experiments show rates up to 1011 Hz with 0.6 pm uncertainty, improving to 0.12 pm at 100 Hz and 64.9 fm at 2.63 Hz with increased averaging, achieving lower tracking uncertainty than recent DOFC-based FBG interrogation systems (1.7 pm [11], 190 fm [12]) while maintaining a compact and cost-effective implementation, making it well suited for high-resolution FBG interrogation in applications requiring fast response.

Acknowledgements

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